



Final Report

GUJCOST-ROBOFEST 5.0

Aerial Robotics: Minefield Navigation Challenge Senior Category



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1. Description of Robot

1.1. Abstract

This project presents an autonomous drone swarm system designed to assist safe navigation across mine-affected terrain. The proposed system consists of five quadcopter drones operating in a coordinated leader–follower architecture. The leader drone interprets human hand gestures using a machine learning–based vision system and directs the swarm accordingly. Each drone is equipped with dual cameras: a front camera for gesture recognition and a downward-facing AI vision camera for terrain monitoring and mine detection. The drones communicate using a WiFi-based swarm communication network and collaboratively explore the environment to identify potential hazards and determine a safe navigation path. The system also incorporates autonomous safety mechanisms such as return-to-home functionality and independent drone operation in case of malfunction. The proposed solution demonstrates the effectiveness of swarm robotics and computer vision techniques for hazardous environment exploration, disaster response, and humanitarian minefield navigation.

Keywords: Swarm Robotics, Autonomous Drones, Gesture Recognition, Mine Detection, Path Planning, Human-Drone Interaction

1.2. Introduction

Minefields and unexploded ordnance remain significant threats in regions affected by conflict and disaster. Traditional mine detection methods often require manual inspection, which exposes personnel to serious risks and slows down exploration of hazardous areas. Autonomous aerial robotics systems provide a safer and more efficient approach to exploring such environments.

The objective of this project is to develop a swarm of autonomous drones capable of detecting potential mines, mapping hazardous terrain, and guiding a human safely through the area. The system utilizes gesture-based control, allowing the operator to interact with the drone swarm without using a remote controller. A machine learning model processes hand gestures captured through a camera mounted on the drone, translating them into navigation commands.

The drones collaborate using a wireless communication network, allowing them to coordinate their movements and share detected hazard information. Through this coordinated exploration, the system identifies safe paths across the minefield and assists the human operator in crossing the area safely.

1.3. System Overview

The proposed system is a **multi-drone aerial robotic platform** designed for safe minefield mapping and human navigation through coordinated autonomous operation. The system comprises multiple quadrotor drones that work collaboratively to survey the operational area, detect mine-like objects, and compute a safe traversal path. The system detects simulated surface-level mine markers as defined in the ROBOFEST 5.0 guidelines to validate swarm coordination and safe path planning.

Each drone is built on a **customized quadrotor frame** optimized for payload integration, vibration reduction, and flight stability. The mechanical design ensures proper center-of-mass alignment, enabling stable low-altitude flight required for visual detection tasks.

All drones are equipped with a **Mini Pix flight controller**, responsible for low-level flight stabilization and control. A **heterogeneous onboard computing architecture** is adopted across the fleet. The **Leader Drone** is equipped with a **Raspberry Pi 5**, which handles computationally intensive tasks such as centralized data fusion, AI-based gesture recognition, and global path planning. The **Follower and Swarming Drones** are equipped with **Raspberry Pi 4**, which perform lightweight perception, contour-based detection, and communication tasks.

The propulsion system consists of **brushless DC motors** driven by **Cyclone 45 A electronic speed controllers (ESCs)**, ensuring smooth thrust response and stable maneuvering. During mine detection, the drones operate at a low altitude of approximately **2 meters**, where the **Raspberry Pi Camera Module 3** provides sufficient ground coverage and resolution for accurate visual detection.

Through collaborative exploration and real-time data sharing, the swarm identifies hazardous locations and determines a safe path that the human operator can follow.

2. Salient Features of Robot

The proposed multi-drone system incorporates several advanced features that enable autonomous hazard detection and safe navigation:

- **Gesture-Based Drone Control:** Human interaction with the drone swarm using vision-based hand gesture recognition.
- **Multi-Drone Swarm Coordination:** Five drones operate collaboratively using a leader–follower architecture.
- **Distributed Mine Detection:** Swarming drones scan separate regions simultaneously for efficient area coverage.
- **Custom Telemetry Communication System:** A team-developed communication framework enables low-latency data exchange between drones.
- **Real-Time Hazard Mapping:** Detected mine locations are georeferenced and integrated into a global map.
- **Autonomous Safe Path Planning:** The Leader Drone computes an optimal path using an A* search algorithm.
- **Mission Monitoring Dashboard:** A web-based interface visualizes drone positions, mine detections, and safe paths in real time.

3. Description of Mechanical Design

The drone platform uses a **custom lightweight quadcopter frame** constructed from carbon fibre with dimensions of **300 × 300 mm**. The structure provides a high strength-to-weight ratio while maintaining stability during flight.

Each drone is equipped with **four brushless DC motors** mounted at the ends of the frame arms. The motors are fitted with **Gemfan Hurricane propellers** optimized for high thrust efficiency and responsive maneuvering.

The structural design accommodates mounting points for onboard electronics, including the flight controller, companion computer, cameras, GPS module, and power distribution board.

Additional **3D-printed mounts** were fabricated to securely attach Raspberry Pi modules and camera units while maintaining optimal sensor alignment.

4. Description of All Logical Steps in the Working of Mechanical Design

The mechanical operation of the drone system follows the following sequence:

1. The quadcopter frame supports all electronic and propulsion components.
2. Brushless DC motors mounted on the frame arms generate thrust.
3. Propellers convert motor rotation into lift force.
4. The Electronic Speed Controllers regulate motor speed according to flight controller commands.
5. Differential motor thrust enables pitch, roll, and yaw motion of the drone.
6. Controlled thrust adjustments allow stable flight, hovering, and navigation during scanning operations.

5. Software Used & Developed

This section describes the software stack used to implement perception, communication, coordination, and control in the proposed multi-drone system. The software architecture follows a distributed sensing and centralized decision-making model, with task allocation based on drone roles.

The software architecture follows a distributed sensing model with centralized mission coordination executed by the Leader Drone.

5.1 Software Stack Overview

Software / Framework	Purpose	Usage in System
ROS (Robot Operating System)	Middleware for inter-process communication	Used on all drones for modular sensor handling, message passing, and task coordination.

Python	Primary programming language	Used for onboard perception, sensor fusion, communication handling, path-planning logic, and backend API development.
OpenCV	Computer vision library	Performs image preprocessing, contour extraction, geometric filtering, and temporal validation for simulated surface-level mine marker detection.
MediaPipe	Vision-based hand tracking framework	Captures hand landmarks on the Leader Drone for gesture-based high-level command recognition under controlled conditions.
TensorFlow Lite	Lightweight ML inference framework	Executes optimized gesture recognition models on the Leader Drone for real-time onboard inference.
PyTorch	Model training and prototyping	Used offline for training and prototyping gesture recognition models; not used during onboard flight execution.
MAVLink	Flight controller communication protocol	Enables reliable command and telemetry exchange between the companion computer and the Mini Pix flight controller.
FastAPI (Backend)	High-performance backend framework	Receives telemetry, detection data, and path updates from the drone swarm, manages mission-level data, and exposes REST/WebSocket APIs for real-time visualization.
React.js (Frontend)	Web-based user interface	Implements an interactive dashboard to visualize drone positions, detected simulated mine markers, safe paths, and overall mission status in real time.
Leaflet.js	Lightweight mapping library	Renders the minefield map and drone trajectories in a browser-based environment without reliance on external map services.
SVG Overlays	Vector-based visualization	Used with Leaflet to draw detected simulated mine markers, 1-meter exclusion zones, and dynamically updated safe path corridors with high visual clarity.

5.2 Role-Based Software Allocation

5.2.1 Leader Drone (Raspberry Pi 5)

The Leader Drone executes computationally intensive and centralized tasks, including:

- Sensor data aggregation from all drones
- Global map generation and maintenance
- Gesture-based human command interpretation using MediaPipe + TensorFlow Lite
- Centralized path planning using the A* algorithm
- Mission coordination and command distribution

The higher computational capacity of Raspberry Pi 5 enables real-time AI inference and planning without affecting system responsiveness.

5.2.2 Swarming Drones (Raspberry Pi 4)

The Swarming Drones perform distributed sensing and local processing tasks:

- Image acquisition from a downward-facing camera
- Preprocessing and contour-based mine detection using **OpenCV**
- Temporal validation of detected objects
- Conversion of image coordinates to world coordinates using EKF outputs
- Transmission of validated detection data to Leader Drone

5.2.3 Follower Drone (Raspberry Pi 4)

The Follower Drone executes navigation and guidance tasks:

- Reception of waypoint-based navigation commands from Leader Drone
- Local position monitoring using EKF-based localization
- Execution of motion commands via the flight controller
- Continuous status and position feedback to the Leader Drone

5.3 Communication Architecture

The system employs a **hybrid communication architecture**, combining wired and wireless links to ensure reliable and low-latency operation.

5.3.1 Companion Computer to Flight Controller Communication

Communication between the **Raspberry Pi (companion computer)** and the **Mini Pix flight controller** is established via a **USB serial connection** using the **MAVLink protocol**.

Functionality includes:

- Transmission of velocity, position, and waypoint commands
- Reception of telemetry data (IMU, GPS, attitude)

- Arming, mode switching, and safety monitoring

This wired communication ensures deterministic and reliable command execution with minimal latency.

5.3.2 Inter-Drone Communication

Inter-drone communication is implemented using a **Wi-Fi-based UDP protocol**, enabling low-latency data exchange at approximately **5–10 Hz**.

The telemetry communication framework used in this system has been **fully developed and implemented by the team**, enabling efficient swarm coordination without relying on off-the-shelf telemetry modules.

Data packets are transmitted in **lightweight JSON format**, reducing overhead and enabling rapid synchronization across the drone fleet.

Data Exchanged	Source	Destination
Mine coordinates & confidence	Swarming Drones	Leader Drone
Global map & safe path	Leader Drone	Follower Drone
Position & status updates	All Drones	Leader Drone

5.3.3 Communication Robustness

UDP is selected to minimize latency and support real-time coordination. Periodic heartbeat and status messages allow the Leader Drone to monitor drone availability. In the event of temporary packet loss, the system continues to operate using the most recent valid data, ensuring mission continuity.

5.4 Path Planning Software

Safe path planning is implemented on the Leader Drone using an iterative A* search algorithm operating on the global hazard map generated from swarm detections. The algorithm operates on the centralized global map generated from swarm detections and computes an optimal traversal path while maintaining a predefined safety buffer around detected mine-like objects. Path updates are triggered dynamically when new hazards are detected, ensuring adaptive navigation.

5.5 Software Design Justification

A distributed software architecture with centralized planning is adopted to balance computational load, power consumption, and scalability. Lightweight perception and communication tasks are

assigned to Swarming and Follower Drones, while AI-based inference and global decision-making are executed exclusively on the Leader Drone.

This design ensures efficient utilization of onboard resources while maintaining synchronized and reliable multi-drone operation.

6. Electronic Components and their working

The electronic architecture of the proposed multi-drone system is designed to support fully onboard autonomous operation without reliance on a ground control station. Each drone is built around a Mini Pix flight controller, responsible for low-level flight stabilization and attitude control. The flight controller continuously processes motion data from the onboard IMU and positional information from the GPS module to maintain stable flight and accurate navigation.

Each drone integrates a companion computer responsible for high-level perception, communication, and mission logic. The Leader Drone is equipped with a Raspberry Pi 5 to handle centralized coordination, gesture recognition, global mapping, and path planning. The Follower and Swarming Drones use Raspberry Pi 4 modules for distributed perception tasks such as terrain scanning, mine detection, and inter-drone communication.

For terrain sensing, a Raspberry Pi Camera Module 3 mounted in a downward-facing configuration captures ground images during low-altitude flight. These images are processed onboard to detect mine-like objects using computer vision techniques. Detected coordinates are fused with EKF-based state estimates and georeferenced to determine the approximate global location of detected hazards.

High-level navigation commands from the companion computer are sent to the flight controller via a serial interface. The flight controller then drives the ESCs to regulate motor speed and maintain flight stability.

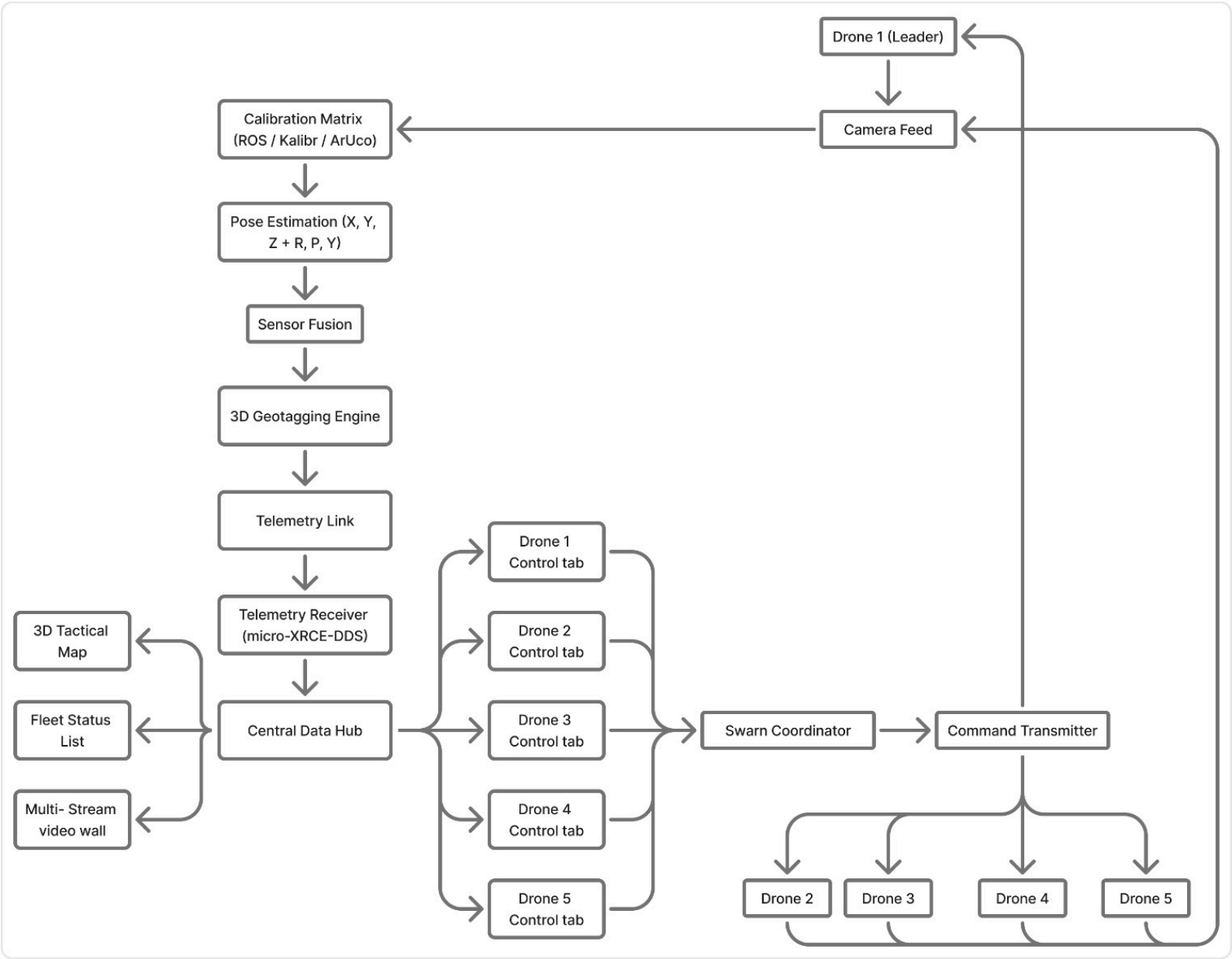
The onboard power distribution system ensures regulated voltage supply to all electronic components while onboard filtering and regulation circuits maintain stable operation during dynamic flight conditions.

6.1. ELECTRONIC COMPONENTS

Component	Role in System	Justification
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Raspberry Pi 5 (16GB)	High-performance processing for AI, Fusion, and Localization Logic.	Uniform, high-memory capacity ensures low-latency and high-throughput processing for centralized data fusion and complex A* path planning.
Raspberry Pi 4	Capture downward frames, detect mines via contours, filter geometrically (area, shape, aspect ratio), and transmit coordinates to Leader Drone via UDP.	Provides sufficient processing capability for real-time image processing and local EKF-based state estimation, while reducing overall system cost and power consumption.
Mini Pix Flight Controller (Radio Link)	Provides low-level flight stabilization and high-quality IMU data.	Established, reliable platform running ArduPilot/PX4 firmware, ensuring predictable flight control and clean sensor data for the companion computer.
Cyclone 45A ESC	Provides robust and precise motor control (45A rating).	A high current rating ensures reliable power management for high-thrust motors, which is crucial for responsive maneuvers and stability under load.
Emax Eco 2004 3000kv Motor	Provides propulsion (3000kv) for high-efficiency flight.	Selected for optimal kV rating, balancing thrust and endurance required for the 15+ minute mission time.
UBEC (Voltage Regulator)	Regulates power supply from the main battery for sensitive electronics.	Provides clean and stable 5V/12V power to companion computers and onboard cameras, preventing voltage fluctuations and brownouts during high-load conditions.
12 MP AI Camera	Provides high-resolution input for Real-Time Gesture Hand Control .	High-resolution (12 MP) and wide field-of-view cameras are required for accurate and responsive detection of human gestures at varying distances.

RPi Camera Module 3	Provides vision input for mine detection.	Optimized for use with Raspberry Pi, delivering sufficient quality for color-based object detection during the 2m altitude sweep.
HGLRC M100 Pro GPS	Provides absolute global position data for georeferencing.	Multi-constellation GNSS provides robust global position locking, serving as the essential absolute correction input for the fusion logic.



Electronic Connection and Swarm System Architecture

8. Operational Workflow of the Final System

The operational workflow of the proposed drone swarm system follows a structured sequence of stages that enable gesture-based human interaction, distributed mine detection, swarm coordination, and safe path guidance. The workflow integrates onboard perception, telemetry communication, centralized mission coordination, and closed-loop feedback to ensure reliable operation of the swarm during the mission.

8.1 System Initialization and Mission Setup

At the start of the mission, all drones are positioned within the designated start zone with propellers disarmed. Upon powering on, each drone performs an internal system health check to verify the functionality of onboard sensors, processing units, and communication interfaces.

During initialization, the **Inertial Measurement Unit (IMU)** and **GPS module** undergo calibration to establish accurate orientation and positional references. The custom telemetry communication system developed by the team is activated, enabling communication between drones and the central mission monitoring interface.

The **Follower Drone is initialized first**, performing an initial reconnaissance flight to determine the operational boundaries. During this flight, the drone records the GPS coordinates of the four corners of the mission area and transmits them to the **Leader Drone**.

Using these coordinates, the Leader Drone defines the mission workspace and partitions the area into sub-regions assigned to individual swarm drones.

The Leader Drone then ascends to a higher altitude to maintain global situational awareness and supervise mission coordination. The remaining **Swarming Drones** are initialized sequentially and begin scanning their assigned regions.

Dynamic workload balancing is implemented to ensure efficient coverage. If a drone finishes early or encounters dense detections, the Leader Drone redistributes remaining tasks to maintain balanced exploration and minimize mission time.

8.2 Localization and Sensor Fusion

Accurate localization is essential for reliable mine mapping and safe navigation. Each drone employs an onboard localization framework based on **sensor fusion techniques**.

The **IMU provides high-frequency orientation and motion data**, while the **GPS module supplies global positional information**. During low-altitude flight, altitude measurements may also be enhanced using additional sensors to improve terrain awareness.

An **Extended Kalman Filter (EKF)** runs onboard each drone to fuse IMU and GPS measurements, producing stable estimates of:

- Position (X, Y, Z)
- Velocity
- Orientation (Roll, Pitch, Yaw)

The EKF reduces sensor noise and drift, allowing the system to accurately estimate drone states and maintain the required **1-meter safety buffer from detected hazards**.

8.3 Vision Processing and Pose Estimation

The vision pipeline begins with image acquisition from onboard cameras. Camera calibration techniques such as **ROS calibration tools, Kalibr, and ArUco-based calibration** are used to compute the calibration matrix required for accurate visual processing.

Using calibrated camera parameters, the system performs pose estimation to determine the drone's spatial orientation relative to the environment. The pose estimation process calculates position and orientation parameters, including:

- X, Y, Z coordinates
- Roll
- Pitch
- Yaw

This spatial information is essential for transforming detected objects from image coordinates into world coordinates for mapping.

8.4 Real-Time Telemetry Communication

The drone swarm uses a custom **telemetry communication framework** developed by the team. This telemetry system enables real-time exchange of mission data between drones and the central monitoring system.

Communication is implemented using a lightweight **UDP-based protocol** operating at approximately **5–10 Hz**, ensuring low latency and efficient bandwidth utilization.

Each drone transmits JSON-formatted packets containing:

- drone position and EKF state estimates
- detected mine coordinates
- sensor readings
- operational status and heartbeat signals

The telemetry data is received by the **Telemetry Receiver**, which forwards the information to the **Central Data Hub** for processing.

This custom telemetry infrastructure enables scalable swarm coordination while maintaining minimal communication overhead.

8.5 Distributed Mine Detection

Mine detection is implemented using a **vision-based contour detection approach**.

Each Swarming Drone is equipped with a downward-facing camera that continuously captures video frames during low-altitude scanning.

The detection pipeline includes:

1. Image Preprocessing

Captured frames undergo preprocessing steps including noise filtering, smoothing, and thresholding to enhance contrast between the terrain and potential mine markers.

2. Contour Extraction

Contours are extracted from the processed images using standard computer vision techniques. Each contour is evaluated using geometric filters such as:

- contour area constraints
- aspect ratio validation
- contour solidity
- boundary smoothness

Contours failing these checks are discarded as false positives.

3. Temporal Validation

To improve reliability, detections are validated across multiple consecutive frames. Only objects that appear consistently across several frames are confirmed as valid mine-like detections.

4. Confidence Estimation

Each detection is assigned a confidence score based on geometric consistency and temporal stability.

8.6 Geotagging and Hazard Mapping

Once a mine-like object is confirmed, its location is transformed into world coordinates using:

- EKF-based drone position
- IMU orientation
- altitude measurements

The resulting coordinates are processed by the **3D Geotagging Engine**, which associates each detection with its spatial location in the global environment.

These geotagged detections are transmitted through the telemetry network and aggregated at the **Central Data Hub**, where redundant detections are removed, and a global mine map is updated.

8.7 Centralized Data Fusion and Swarm Coordination

The **Leader Drone** acts as the **central decision-making unit** of the swarm. It aggregates detection data from all Swarming Drones and integrates it into a centralized hazard map.

The **Swarm Coordinator module** then assigns exploration tasks to each drone to ensure efficient coverage while preventing redundant scanning.

Navigation commands are generated based on mission objectives and hazard information.

8.8 Safe Path Planning

Using the global hazard map, the Leader Drone computes a safe traversal path across the minefield.

A graph-based path planning algorithm is used to generate a route that maintains the mandatory **1-meter safety buffer** from all detected hazards.

The computed path is continuously updated if new hazards are detected or if localization deviations occur.

8.9 Human Guidance and Closed-Loop Feedback

Human interaction is enabled through gesture recognition performed by the **Leader Drone**.

The Leader Drone captures video frames of the human participant and processes them using a lightweight gesture recognition pipeline. Recognized gestures are translated into high-level commands influencing navigation behavior. The **Follower Drone acts as the physical guide**, following waypoint commands generated by the Leader Drone and guiding the human along the safe path.

The system operates in a **closed-loop feedback architecture**, continuously updating navigation commands based on sensor data, detection results, and drone states. If a new hazard is detected during traversal, the system dynamically recomputes the safe path and updates navigation commands.

This adaptive feedback loop ensures safe and reliable mission execution.

Data Flow Architecture

The proposed drone swarm system follows a structured data processing pipeline that integrates visual perception, sensor fusion, geotagging, telemetry communication, and swarm coordination.

The data flow begins with image acquisition from the onboard cameras mounted on the drones. Visual frames captured by the cameras undergo preprocessing and calibration using tools such as ROS calibration utilities, Kalibr, and ArUco marker detection. These calibration procedures generate the camera calibration matrix required for accurate spatial interpretation of the environment.

Following calibration, the system performs pose estimation to determine the drone's spatial orientation. Pose estimation calculates position and orientation parameters including X, Y, Z coordinates along with roll, pitch, and yaw angles. This information enables accurate transformation of detected objects from image coordinates to world coordinates.

Sensor fusion is then performed using data from the IMU and GPS modules. An Extended Kalman Filter integrates these measurements to produce a reliable estimate of the drone's position, velocity, and orientation while reducing sensor noise and drift.

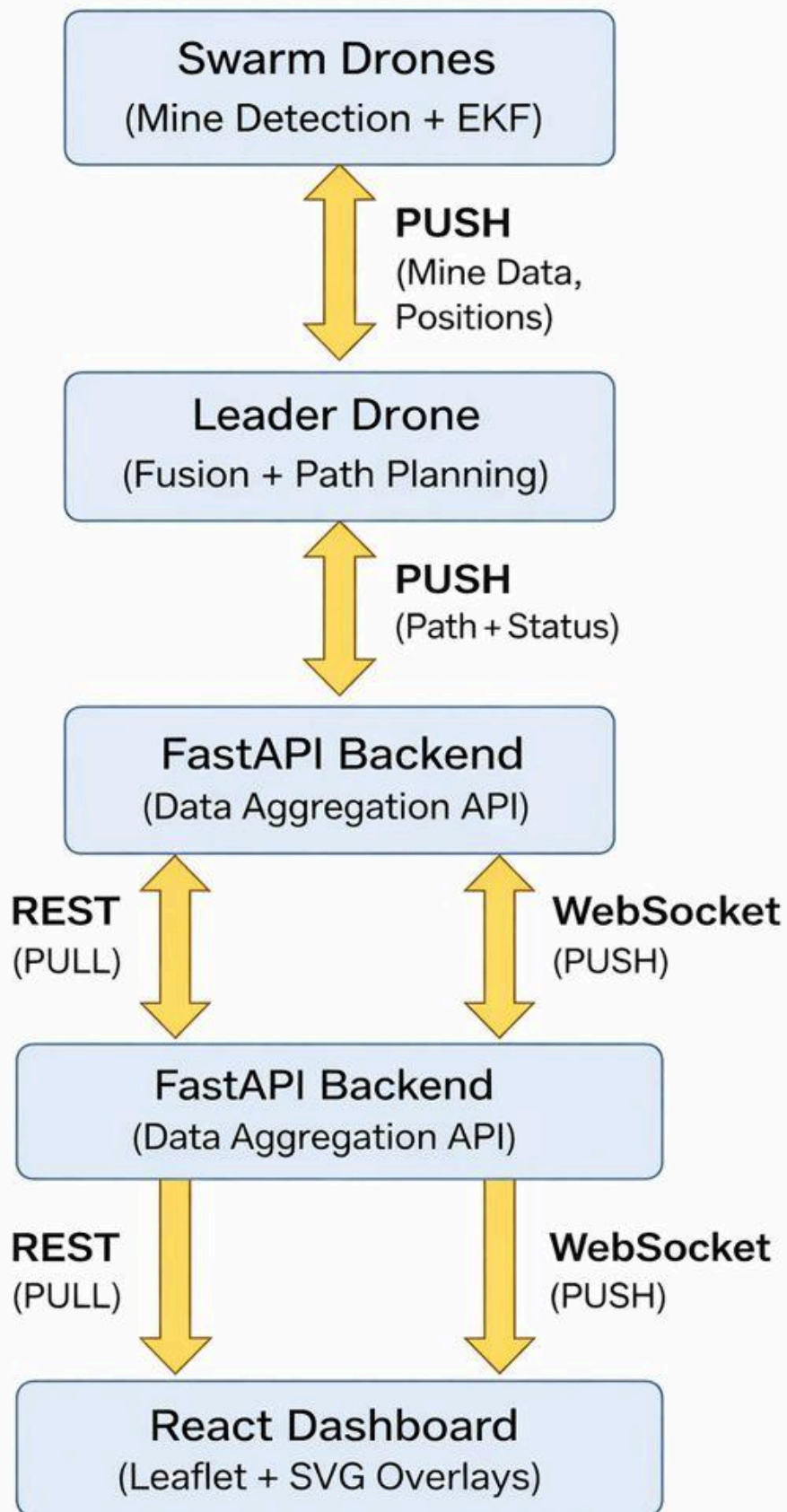
Once potential hazards are detected by the vision system, their coordinates are processed through the **3D geotagging engine**, which associates each detection with its corresponding spatial location in the global environment.

The processed detection data, along with drone state information, is transmitted through the **custom telemetry communication system** using a lightweight UDP-based communication protocol. The telemetry receiver forwards the incoming data to the Central Data Hub, where information from all drones is aggregated.

The **Central Data Hub** distributes this information to the **Swarm Coordination Module**, which analyzes the global hazard map and generates navigation commands for each drone in the swarm.

These commands are transmitted back to the drones through the **command transmitter module**, enabling coordinated swarm movement and adaptive mission execution.

Through this structured data pipeline, the system achieves reliable hazard detection, efficient swarm coordination, and real-time mission monitoring.



Dashboard Data Flow

9. Futuristic Application of Robot and Commercialization

Future improvements may include integration of advanced deep-learning-based object detection models for improved hazard detection accuracy, enhanced swarm coordination algorithms, and extended operational range for large-scale minefield exploration.

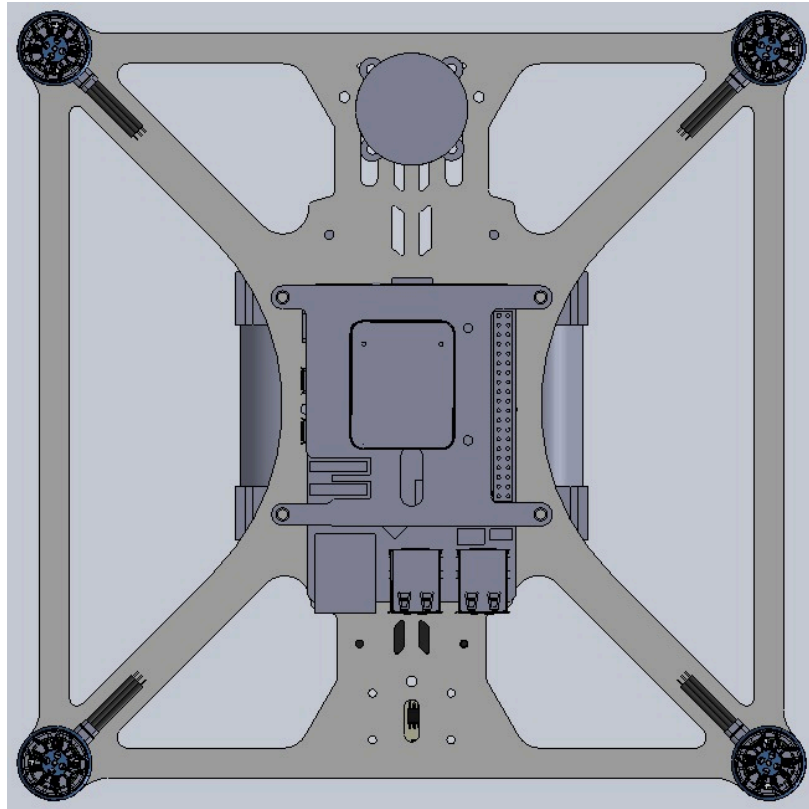
The proposed drone swarm system has potential applications in several domains:

- **Humanitarian Demining:** Safe detection and mapping of landmines in post-conflict regions.
- **Disaster Response:** Exploration of hazardous environments such as earthquake-damaged areas.
- **Military Reconnaissance:** Autonomous terrain monitoring and hazard detection.
- **Search and Rescue Operations:** Rapid scanning of disaster zones.
- **Industrial Safety Monitoring:** Inspection of hazardous industrial environments.

With further development, the system could evolve into a **commercial autonomous hazard-mapping platform** for defense and disaster-management agencies.

APPENDIX

Mechanical Design and CAD Model



PHOTOS



Links for Video.

<https://drive.google.com/drive/folders/1Zn6hnmbidfBRnTnsvFy0fLmwMWv67bxV?usp=sharing>

SIGNED BY-

TEAM MEMBERS

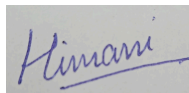
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